

The Action Potential

“Spikes”

Computational Neuroscience

University of Bristol

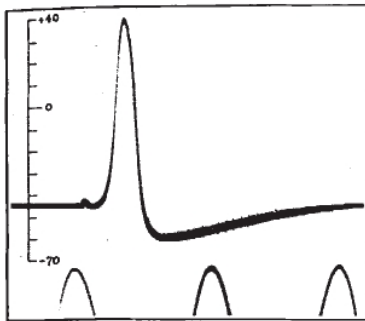
M Rule

Learning outcomes:

- ▶ Describe what an action potential is, what challenges it solves, etc.
- ▶ Describe the phases of an action potential, and the roles of various ionic currents and channels in each phase
- ▶ Distinguish mechanisms of relative vs. absolute refractory period

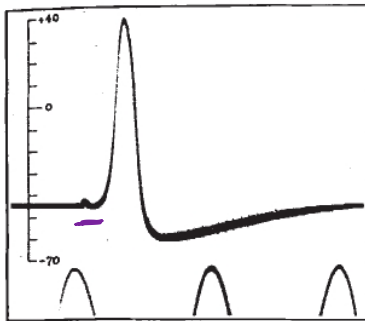
Spikes

Neurons use spikes for reliable long-range communication



Oscilloscope trace from *Action Potentials Recorded from Inside a Nerve Fibre*, Hodgkin and Huxley (1939): ***“Action potential recorded between inside and outside of axon. Time marker 500 cycles/sec. The vertical scale indicates the potential of the internal electrode in millivolts, the sea water outside being taken at zero potential.”***

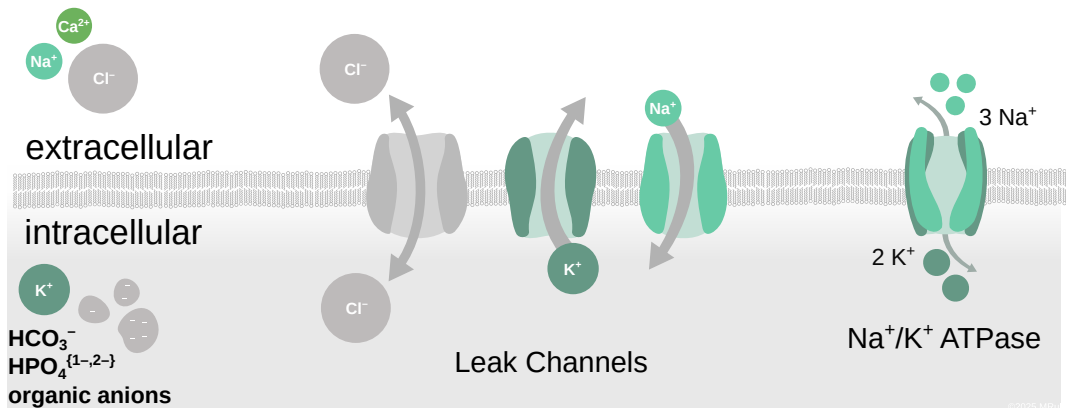
Neurons use spikes for reliable long-range communication



The action potential (“spike”) is a sharp rise and fall in cell’s membrane voltage that’s produced when the cell exceeds a **threshold voltage**.

All-or-nothing, binary $\{0, 1\}$

- Robust: regenerative wave that can travel long distances along axons/dendrites without decay



The Action Potential

Initiation Voltage-gated Na^+ channels have a threshold of ~ -55 mV; If the v_m exceeds this, Na^+ channels open. Na^+ flows *into* cell, pulling v_m toward $E_{\text{Na}^+} \sim +60$ mV

Depolarisation The cell spikes if positive feedback from voltage-gated Na^+ channels overwhelms other currents: v_m , crossing 0 mV and reaching a peak at $\sim +30$ – 40 mV

Peak v_m stops rising because the depolarisation...

- ▶ Inactivates voltage-gated Na^+ channels (Na^+ current stops)
- ▶ Opens voltage-gated K^+ channels (threshold ~ -40 – -30 mV), leading to ...

Repolarisation K^+ efflux pulls v_m toward $E_{\text{K}^+} \sim -95$ mV;

Hyperpolarization Voltage-gated K^+ channels close slowly, continue to pull cell below v_{rest} to ~ -80 – -75 mV.

Refractory period

Absolute

- ▶ ~ 1–2 ms
- ▶ Enough voltage-gated Na^+ channels are inactive that no physiological input can trigger a new spike

Relative

- ▶ ~ 2–5 ms
- ▶ Voltage-gated Na^+ channels have started to de-inactivate
- ▶ Some voltage-gated K^+ channels remain open
- ▶ Cell remains hyperpolarized, and requires a strong input to spike.

Exam confusions

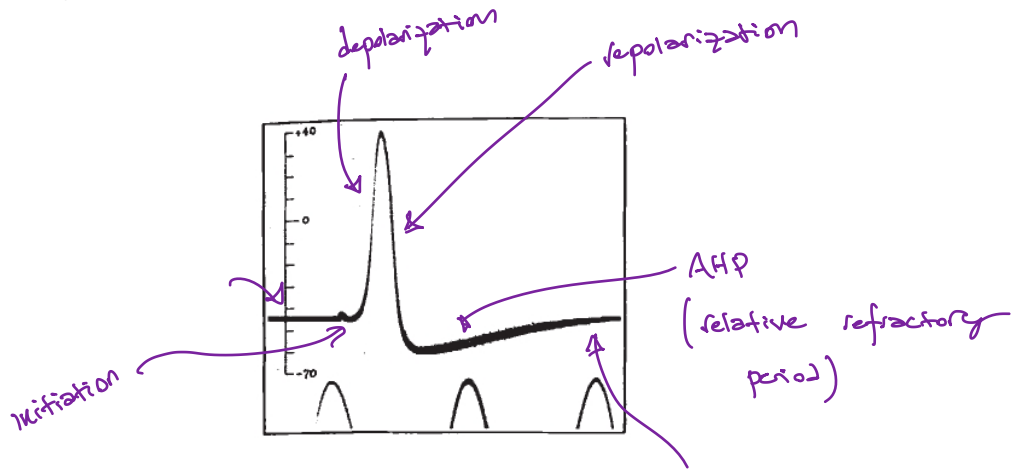
The Na^+/K^+ ATPase is *not* immediately required to repolarise the neuron after an action potential. Each spike uses a small amount of the electrical potential stored in Na^+ and K^+ gradients.

Exam confusions

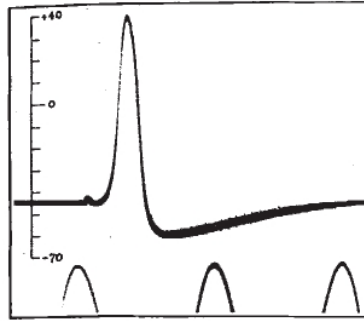
Try not to confuse

- 1 *voltage-gated* Na^+ and K^+ channels involved in the *action potential*
- 2 *ligand-gated* ionotropic neurotransmitter receptors
(synapse: chemical $\rightarrow$ electrical)
- 3 *voltage-gated* Ca^{2+} channels in axonal terminals
(synapse: electrical $\rightarrow$ chemical)

label the phases of this action potential



what voltage gates and ion fluxes are involved in each phase?



Draw an action potential from member, including realistic voltages ± 15 mV, and label its key phases

More realism

Action potentials *travel* out along axons

- ▶ Regenerative soliton wave
- ▶ Refractory period ensures directed propagation

Cable equation

Saltatory conduction

- ▶ Myelin
- ▶ Nodes of Ranvier

Neural computation and communication is implemented via electrochemical reactions

Signal transmission via spikes

- 1 ***Spike initiation and propagation:*** via voltage-gated ion channels
- 2 ***Pre-synaptic:*** axonal terminal depolarises → neurotransmitter release
- 3 ***Post-synaptic:*** chemical signals → electrical currents in the dendrite
- 4 (repeat)

end

